HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and validated **Python** statistical programming code for the analysis and reporting of clinical trial data, ensuring adherence to industry standards.
- Collaborated with **data scientists** and statisticians to define data requirements and analytical needs for complex clinical trial datasets.
- Created and maintained comprehensive **documentation** for programming processes, code repositories, and analytical outputs to support regulatory compliance.
- Participated in detailed reviews of analysis datasets and final clinical trial reports, verifying accuracy and completeness of findings.
- Built and maintained **knowledge graphs** using Python to represent relationships within clinical trial data, enhancing analytical capabilities.
- Implemented **AI tools and technologies** to improve the efficiency and depth of clinical data analysis, supporting research objectives.
- Deployed Python-based AI solutions on **AWS**, managing infrastructure and ensuring scalability for large-scale data processing.

SKILLS

Programming Languages – Python, SQL, R

Cloud Platforms – AWS (EC2, S3, Lambda, CloudWatch)

AI & Data Science – AI tools and technologies, Data Engineering, Data Science, Advanced Analytics, Statistical Programming, Knowledge Graphs, Machine Learning Fundamentals

Tools & Technologies – Docker, Git, Jenkins, Jira, Confluence

Databases – PostgreSQL, MySQL, Oracle

Methodologies – Agile, Scrum

PROJECTS

Clinical Data Analysis Enhancement Platform

- Developed a Python-based platform to streamline the analysis of clinical trial data, integrating various statistical libraries for enhanced reporting.
- Implemented automated data validation routines that reduced manual review time by approximately **40%** for key analysis datasets.
- Designed and deployed the platform on **AWS Lambda**, ensuring efficient processing of large datasets and on-demand scalability.

Knowledge Graph for Clinical Trial Insights

- Engineered a **knowledge graph** using Python to map complex relationships between patient demographics, treatment protocols, and trial outcomes.
- Extracted and transformed data from multiple sources, populating the graph with over **50,000** interconnected data points.
- Facilitated data exploration for clinical researchers, enabling quicker identification of potential correlations and insights.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)